

Aaditya Dawar

1024150093

2013

Assignment-9: File Management

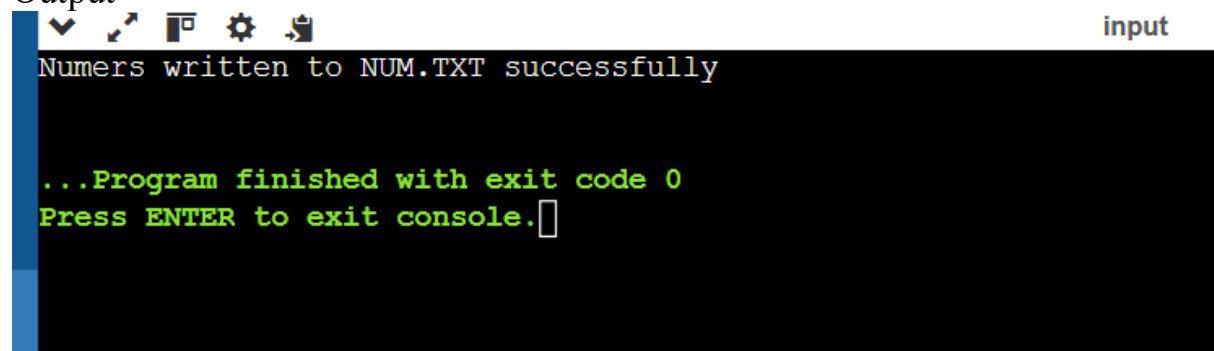
1. Write a C++ program to write number 1 to 200 in a data file NUM.TXT.

Code-

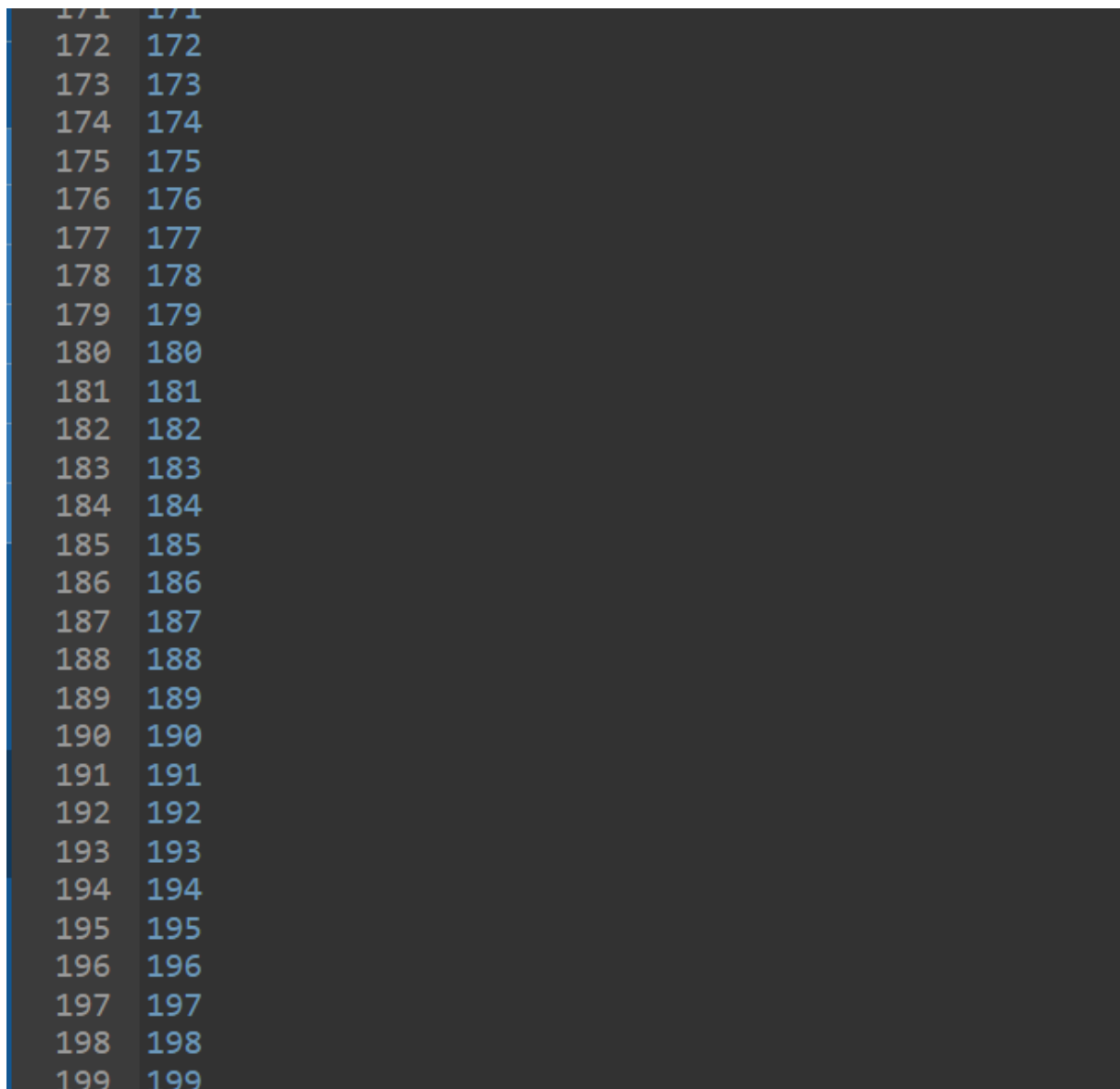
```
#include <iostream>
#include <fstream>
using namespace std;
int main(){
    ofstream file;
    file.open ("NUM.TXT");

    if (!file){
        cout<<"File did not open successfully"<<endl;
    }
    for (int i=1;i<201;i++){
        file<<i<<endl;
    }
    file.close();
    cout<<"Numers written to NUM.TXT successfully"<<endl;
    return 0;
}
```

Output-

A screenshot of a console window with a black background and green text. The window has a title bar with standard icons and the word "input" on the right. The output text reads: "Numers written to NUM.TXT successfully", followed by "...Program finished with exit code 0" and "Press ENTER to exit console." with a cursor.

```
input
Numers written to NUM.TXT successfully
...Program finished with exit code 0
Press ENTER to exit console.
```



2. Write a user-defined function in C++ to read the content from a text file NOTES.TXT, count and display the number of alphabets present in it.

Code- #include <iostream>

#include <fstream>

using namespace std;

int countalphabets(){

 ifstream file("NOTES.TXT");

 if (!file){

 cout << "File did not open" << endl;

 return 0;

 }

```

char ch;
int count = 0;

while(file.get(ch)){
    if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')){
        count++;
    }
}

file.close();
return count;
}

int main(){
    int total = countalphabets();
    cout << "Number of alphabets = " << total << endl;
    return 0;
}

```

OUTPUT-

```

File did not open
Number of alphabets = 0

...Program finished with exit code 0
Press ENTER to exit console.

```

3. Write a program to perform read/write binary I/O operation on a file (i.e. write the object of the class to a file)

Code-

```

#include <iostream>
#include <fstream>
using namespace std;
class Student {
    int roll;
    char name[20];
public:
    void input() {
        cout << "Enter roll number: ";
        cin >> roll;
    }
}

```

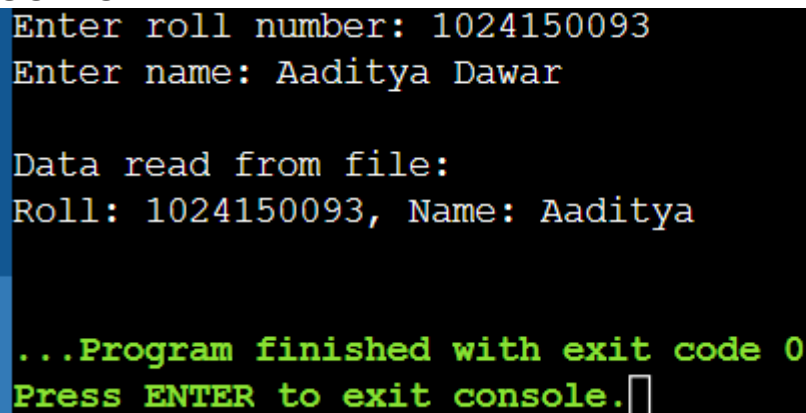
```

        cout << "Enter name: ";
        cin >> name;
    }
    void display() {
        cout << "Roll: " << roll << ", Name: " << name << endl;
    }
};

int main() {
    Student s;
    ofstream fout("STUDENT.DAT", ios::binary);
    s.input();
    fout.write((char*)&s, sizeof(s));
    fout.close();
    ifstream fin("STUDENT.DAT", ios::binary);
    fin.read((char*)&s, sizeof(s));
    cout << "\nData read from file:\n";
    s.display();
    fin.close();
    return 0;
}

```

OUTPUT-



```

Enter roll number: 1024150093
Enter name: Aaditya Dawar

Data read from file:
Roll: 1024150093, Name: Aaditya

...Program finished with exit code 0
Press ENTER to exit console.

```

4. Write a program to copy the contents of one file to another.

Code-

```

#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream fin("NUM.TXT");
    ofstream fout("TARGET.TXT");

    if (!fin || !fout) {

```

```

        cout << "Error opening file!" << endl;
        return 1;
    }

    char ch;

    while (fin.get(ch)) {
        fout.put(ch);
    }

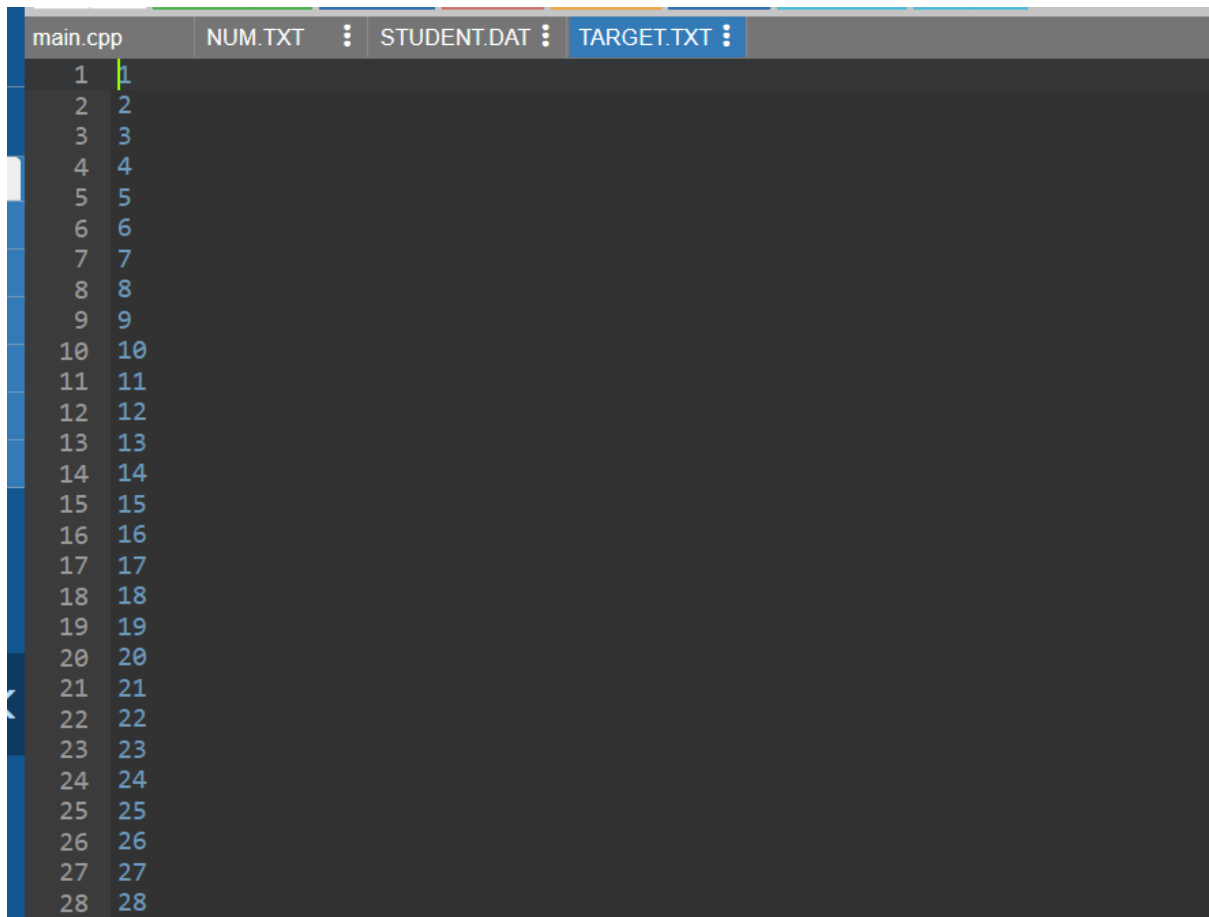
    fin.close();
    fout.close();

    cout << "File copied successfully." << endl;
    return 0;
}

```

OUTPUT-

The screenshot shows a terminal window with the output of a C++ program. The first line is "File copied successfully." in yellow text on a black background. The second line is "...Program finished with exit code 0" in green text. The third line is "Press ENTER to exit console" in green text, followed by a cursor. Below the terminal window is a code editor window with a dark background. The editor shows a list of files: "main.cpp", "NUM.TXT", "STUDENT.DAT", and "TARGET.TXT". The "main.cpp" file is selected, and its contents are displayed in a list of lines, numbered 171 to 200. The lines are: 171 171, 172 172, 173 173, 174 174, 175 175, 176 176, 177 177, 178 178, 179 179, 180 180, 181 181, 182 182, 183 183, 184 184, 185 185, 186 186, 187 187, 188 188, 189 189, 190 190, 191 191, 192 192, 193 193, 194 194, 195 195, 196 196, 197 197, 198 198, 199 199, and 200 200.



5. Write a program to implement I/O operations on characters. I/O operations include inputting a string, calculating the length of the string, storing the string in a file and fetching the stored characters from it.

Code-

```
#include <iostream>
#include <fstream>
using namespace std;
```

```
int main() {
    char str[100];
    cout << "Enter a string: ";
    cin.getline(str, 100);
    int length = 0;
    for (int i = 0; str[i] != '\0'; i++) {
        length++;
    }

    cout << "Length of string = " << length << endl;

    ofstream fout("DATA.TXT");
```

```

if(!fout) {
    cout << "Error opening file!" << endl;
    return 1;
}

fout << str;
fout.close();

ifstream fin("DATA.TXT");
if(!fin) {
    cout << "Error opening file!" << endl;
    return 1;
}

char ch;
cout << "\nData read from file:\n";

while (fin.get(ch)) {
    cout << ch;
}

fin.close();

return 0;
}

```

OUTPUT-

```

Enter a string: aadityadawar
Length of string = 12

Data read from file:
aadityadawar

...Program finished with exit code 0
Press ENTER to exit console.

```

main.cpp	NUM.TXT	STUDENT.DAT	TARGET.TXT	DATA.TXT
1	aadityadawar			

6. (a) Create a file A-Z and read 10th character using seekg().

Code-

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream fout("file.txt");
    for(char ch = 'A'; ch <= 'Z'; ch++) {
        fout << ch;
    }
    fout.close();

    ifstream fin("file.txt");
    fin.seekg(9, ios::beg);

    char ch;
    fin.get(ch);

    cout << "10th character is: " << ch;

    fin.close();
    return 0;
}
```

OUTPUT-


```
10th character is: J

...Program finished with exit code 0
Press ENTER to exit console.□
```

```
main.cpp  file.txt  ⋮
1  ABCDEFGHIJKLMNOPQRSTUVWXYZ
```

(b) Overwrite 5th number using seekp().

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
int main() {
```

```
    ofstream fout("file.txt");
```

```
    for(char ch = 'A'; ch <= 'Z'; ch++) {
```

```
        fout << ch;
```

```
    }
```

```
    fout.close();
```

```
    char newChar;
```

```
    cout << "Enter character to overwrite at 5th position: ";
```

```
    cin >> newChar;
```

```
    fstream file("file.txt", ios::in | ios::out);
```

```
    file.seekp(4, ios::beg);
```

```

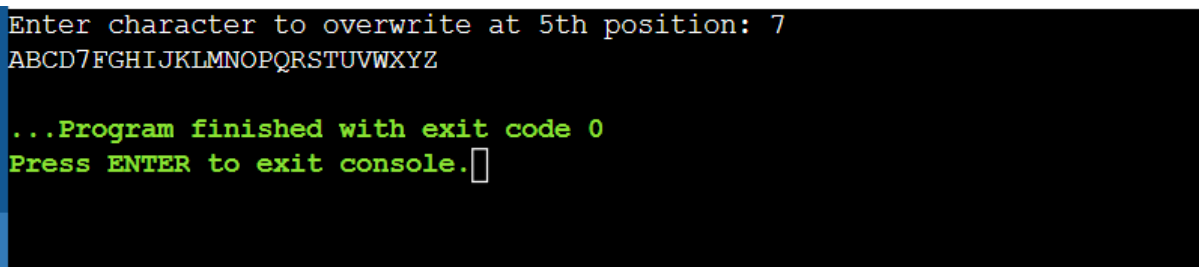
        file.put(newChar);

    file.close();

    ifstream fin("file.txt");
    char ch;
    while(fin.get(ch)) {
        cout << ch;
    }
    fin.close();

    return 0;
}

```



```

Enter character to overwrite at 5th position: 7
ABCD7FGHIJKLMNOPQRSTUVWXYZ

...Program finished with exit code 0
Press ENTER to exit console.

```

(c) Find file size using tellg().

```

#include <iostream>

#include <fstream>

using namespace std;

int main() {
    ofstream fout("file.txt");

    for(char ch = 'A'; ch <= 'Z'; ch++) {

```

```

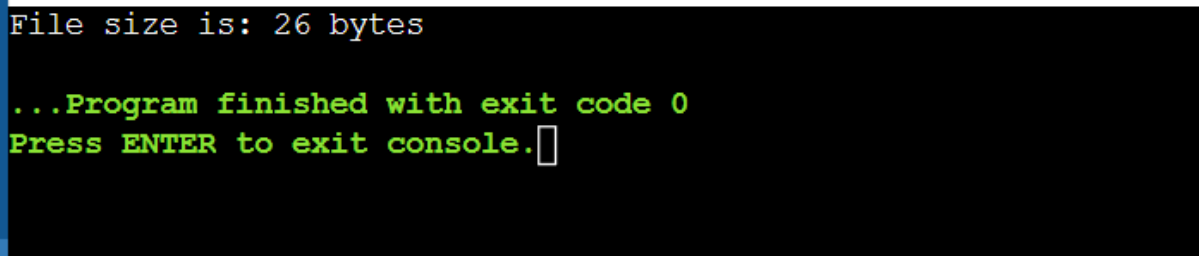
        fout << ch;
    }
    fout.close();

    ifstream fin("file.txt");
    fin.seekg(0, ios::end);
    int size = fin.tellg();

    cout << "File size is: " << size << " bytes";

    fin.close();
    return 0;
}

```



```

File size is: 26 bytes
...Program finished with exit code 0
Press ENTER to exit console.

```

(d) Read last character using seekg().

```

#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream fout("file.txt");

    for(char ch = 'A'; ch <= 'Z'; ch++) {
        fout << ch;
    }
}

```

```

    }
    fout.close();

    ifstream fin("file.txt");
    fin.seekg(-1, ios::end);

    char ch;
    fin.get(ch);

    cout << "Last character is: " << ch;

    fin.close();
    return 0;
}

```



```

Last character is: z
...Program finished with exit code 0
Press ENTER to exit console.

```

(e) Create a file data.txt with multiple lines. Read the file and:

- i. Move the pointer to the 10th byte using seekg()
- ii. Display the current position using tellg().
- iii. Read and display the remaining content.

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
int main() {
```

```
ofstream fout("data.txt");  
fout << "This is line one.\n";  
fout << "This is line two.\n";  
fout << "This is line three.\n";  
fout.close();
```

```
ifstream fin("data.txt");
```

```
fin.seekg(9, ios::beg);
```

```
cout << "Current position: " << fin.tellg() << endl;
```

```
char ch;
```

```
while(fin.get(ch)) {
```

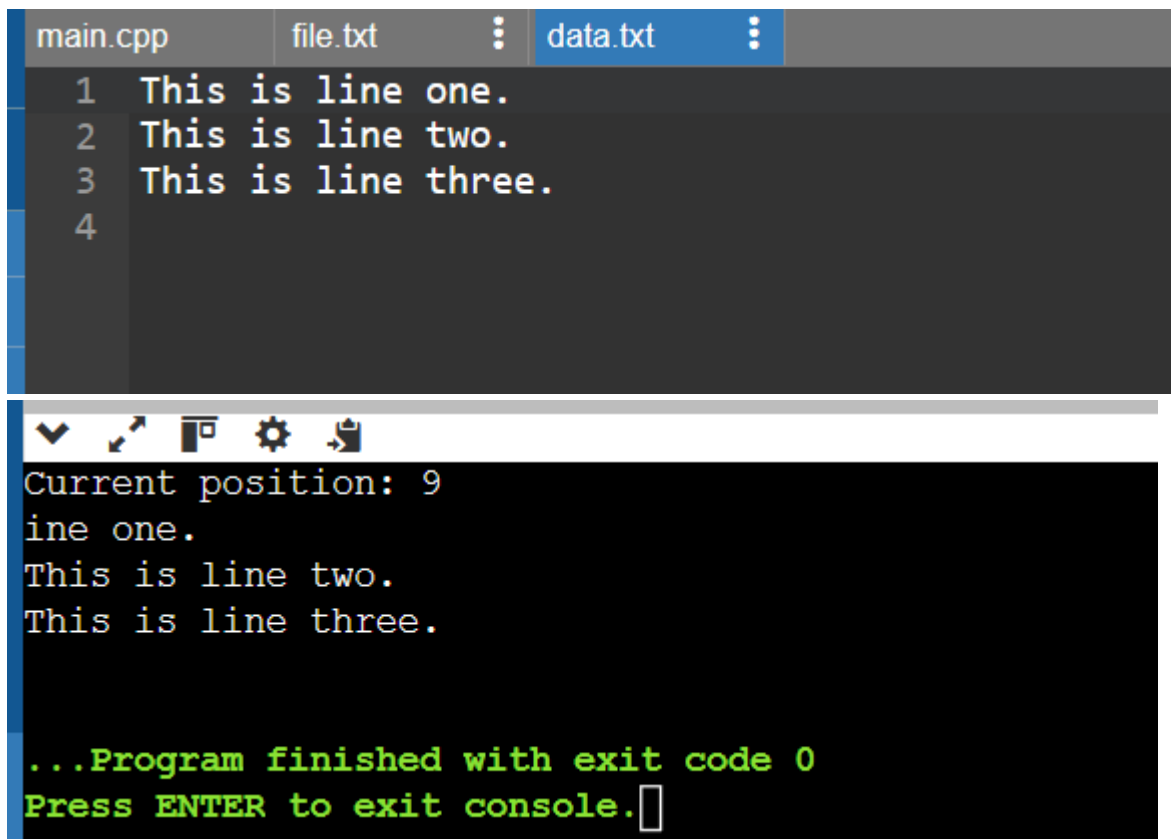
```
    cout << ch;
```

```
}
```

```
fin.close();
```

```
return 0;
```

```
}
```



```
main.cpp  file.txt  data.txt
1 This is line one.
2 This is line two.
3 This is line three.
4
Current position: 9
ine one.
This is line two.
This is line three.
...Program finished with exit code 0
Press ENTER to exit console.
```

(f) Write a program to:

- Create a file with text "HelloWorld"
- After writing each character, display the current position of the put pointer using `tellp()`.
- Replace "World" with "C++" using random access

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
int main() {
```

```
    fstream file("data.txt", ios::out | ios::in | ios::trunc);
```

```
    string str = "HelloWorld";
```

```
    for(char ch : str) {
```

```
    file.put(ch);  
    cout << file.tellp() << endl;  
}  
  
file.seekp(5, ios::beg);  
file.write("C++", 3);  
  
file.close();  
  
ifstream fin("data.txt");  
char ch;  
while(fin.get(ch)) {  
    cout << ch;  
}  
fin.close();  
  
return 0;  
}
```



input

```
1
2
3
4
5
6
7
8
9
10
```

```
HelloC++ld
```

```
...Program finished with exit code 0
```

```
Press ENTER to exit console.
```